

GAP EROSION AND ALL-TIME FIBRES FOR RANK-COMPRESSION SUPPORT DYNAMICS

ANONYMOUS

ABSTRACT. Order a subset $A = \{a_0 < \dots < a_{k-1}\}$ of $\{0, \dots, n-1\}$, subtract the rank j from a_j , discard multiplicity, and iterate the resulting support map. We prove that at time t each original consecutive gap g has become $g - t$ if positive and has otherwise disappeared. This gives an exact pointwise clock, all basins, and the unique state of depth $n - 1$. A nonempty set B is in the time- t image exactly when $\max B + t(|B| - 1) < n$. Every such target has a coefficient-sum fibre formula in which arbitrary short-gap words occupy the slots around forced long gaps; at one step it reduces to $\binom{n - \max B}{|B|}$, and the first image has F_{n+2} states. A bounded-gap generating function gives every depth population. Classical strict/weak sequence shifts, stars and bars, beta sets, and Fibonacci summation receive no contribution credit. The retained conjunction is OWNER_AMBER; circulation remains HOLD_EXTERNAL.

1. THE MAP AND THE SUBTRACTION BOUNDARY

Let $[n]_0 = \{0, \dots, n-1\}$ and $\mathcal{X}_n = 2^{[n]_0}$. For $A = \{a_0 < \dots < a_{k-1}\}$ define

$$(1) \quad T_n(A) = \text{supp}\{a_j - j : 0 \leq j < k\}, \quad T_n(\emptyset) = \emptyset,$$

where supp discards multiplicity. The integers $a_j - j$ are weakly increasing and lie in $[n]_0$, so this is a total finite self-map.

The strict-to-weak sequence shift behind $a_j - j$ and the resulting stars-and-bars identities are classical [2]. Beta sets are also standard partition coordinates; see, for example, their use in core partition theory [1]. We assign those encodings, bounded compositions, Fibonacci identities, and generic functional-graph bookkeeping no contribution credit. The scoped residual is only the autonomous support iteration (1), its all-time gap law, and the simultaneous pointwise/image/every-target inverse atlas below. A bounded search found no source stating that literal iterative conjunction. This non-hit is neither novelty nor priority evidence.

Internally, P163 acts on families by complemented shadows, P169 transfers last occurrences between set-partition blocks, and P179 randomly isolates labels through commuting idempotents. None is (1); nevertheless their shadow, partition, and inverse vocabulary is fully subtracted. P100's digit-erasure clock and P178's image-tower bookkeeping likewise receive no separation credit.

For nonempty A , put $m(A) = a_0$ and define its positive gap word

$$(2) \quad g(A) = (g_1, \dots, g_{k-1}), \quad g_j = a_j - a_{j-1}.$$

The empty word is allowed. Given a positive word g , let $E_t(g)$ be the subsequence $(g_j - t : g_j > t)$, in the original order.

2. POINTWISE DYNAMICS AND THE SHARP CLOCK

Theorem 1 (all-time gap normal form). *For every $A \in \mathcal{X}_n$ and $t \geq 0$, the empty set remains empty. If A is nonempty, then $\min T_n^t(A) = m(A)$ and*

$$(3) \quad g(T_n^t(A)) = E_t(g(A)).$$

Equivalently, start at $m(A)$ and cumulatively add, in order, the terms $g_j - t$ for which $g_j > t$.

The recurrent states are precisely $\emptyset$ and the n singletons. Writing $\tau(A)$ for entrance time into this fixed set,

$$(4) \quad \tau(A) = \begin{cases} 0, & |A| \leq 1, \\ \max_{1 \leq j < |A|} g_j, & |A| \geq 2. \end{cases}$$

For $n \geq 2$ the global height is $n - 1$, attained uniquely by $\{0, n-1\}$. The basin of $\{m\}$ has size 2^{n-m-1} , and the empty basin has size one.

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Proof. Put $b_j = a_j - j$. Consecutive differences in the weakly increasing list (b_j) satisfy

$$(5) \quad b_j - b_{j-1} = g_j - 1.$$

Taking support deletes exactly the zero differences. Deleting repetitions does not merge positive increments: it retains, in their old order, exactly the positive values $g_j - 1$. The minimum remains $b_0 = a_0$. One application therefore replaces the gap word by $E_1(g)$, and induction proves (3).

If $|A| \geq 2$, all gaps have disappeared at time $\max g_j$, while a gap equal to that maximum is still positive one epoch earlier. This proves the least-time formula (4) and excludes nonfixed recurrence. A gap is at most $n - 1$, with equality only when two consecutive selected elements are $0, n - 1$; no third selected element can then occur. This proves the sharp and unique extremal statement. Finally, a nonempty orbit preserves its minimum and ends at that singleton. The subsets with minimum m are $\{m\} \cup C$ with $C \subseteq \{m+1, \dots, n-1\}$, giving the basin count. $\square$

3. ALL-TIME IMAGES AND TARGET-LOCAL FIBRES

For $t \geq 0$, write

$$(6) \quad S_t(z) = z + \dots + z^t,$$

with $S_0(z) = 0$.

Theorem 2 (complete time- t inverse atlas). *Fix $t \geq 0$. The empty target has the unique source $\emptyset$. A nonempty target $B = \{b_0 < \dots < b_r\}$ occurs at time t if and only if*

$$(7) \quad b_r + tr < n.$$

For every B (with a negative upper limit interpreted as zero),

$$(8) \quad |(T_n^t)^{-1}(B)| = \sum_{s=0}^{n-1-b_0-(b_r-b_0)-tr} [z^s] (1 - S_t(z))^{-(r+1)}.$$

Consequently,

$$(9) \quad |\operatorname{im} T_n^t| = 1 + \sum_{r \geq 0} \binom{n-tr}{r+1},$$

where impossible binomial coefficients vanish. At one step,

$$(10) \quad |T_n^{-1}(B)| = \binom{n - \max B}{|B|} \quad (B \neq \emptyset),$$

$$(11) \quad |\operatorname{im} T_n| = F_{n+2},$$

for $F_0 = 0, F_1 = 1$.

Proof. Let $h_i = b_i - b_{i-1}$ for $1 \leq i \leq r$. By Theorem 1, a source mapping to B at time t has the unique gap factorisation

$$(12) \quad U_0, (h_1 + t), U_1, (h_2 + t), \dots, (h_r + t), U_r,$$

where each U_i is an arbitrary ordered word with letters in $\{1, \dots, t\}$. Conversely every such factorisation yields B : the U_i letters disappear and the distinguished gaps lose t .

The distinguished gaps consume total span

$$\sum_{i=1}^r (h_i + t) = b_r - b_0 + tr.$$

The available span after the invariant minimum b_0 is $n - 1 - b_0$. Each short-gap slot has ordinary generating function $(1 - S_t(z))^{-1}$, so summing its $(r+1)$ -slot coefficients through the remaining span proves (8). With every U_i empty, feasibility is exactly (7); this also proves sufficiency.

For fixed r , condition (7) chooses an $(r+1)$ -set from $\{0, \dots, n - 1 - tr\}$, which proves (9). If $t = 1$, then $S_1(z) = z$ and the hockey-stick identity reduces (8) to (10). Finally, $1 + \sum_{r \geq 0} \binom{n-r}{r+1} = F_{n+2}$, proving (11). $\square$

Formula (8) is target-local and remains informative after the first image: its $r+1$ factors distinguish the short-gap words before, between, and after the long gaps that survive to B . In particular, it is not inferred by dividing the carrier size by the image size.

4. EVERY DEPTH POPULATION

Proposition 3 (bounded-gap depth census). *Let $C_n(h) = |\{A \in \mathcal{X}_n : \tau(A) \leq h\}|$ and put $Q_h(z) = (1 - S_h(z))^{-1}$. For every $h \geq 0$,*

$$(13) \quad C_n(h) = 1 + \sum_{s=0}^{n-1} (n-s)[z^s]Q_h(z).$$

Thus the exact depth- h population is $C_n(h) - C_n(h-1)$ for $h \geq 1$, while $C_n(0) = n+1$.

Proof. Besides the empty set, choose the minimum m and then an arbitrary ordered gap word with letters at most h . Such a word of total span s is counted by $[z^s]Q_h(z)$ and permits exactly $n-s$ minima $m \in \{0, \dots, n-1-s\}$. The clock formula says precisely that these are the states of depth at most h . Summation proves (13); at $h=0$ only the empty gap word remains. $\square$

For orientation, at $n=4$ the arrows include

$$\{0, 3\} \mapsto \{0, 2\} \mapsto \{0, 1\} \mapsto \{0\},$$

which witnesses the sharp height. At $n=18$, exhaustive evaluation gives 6,765 first-image targets, 19 fixed states, maximum one-step fibre 2,002, and the unique depth-17 state $\{0, 17\}$.

5. EXACT CONTROL, LIMITATIONS, AND DECLARATIONS

The paper-local standard-library verifier exhausts all 2^n subsets through $n=18$. It compares literal iteration with (3), checks the least pointwise clock, reconstructs every time- t image and every target fibre through (8), and checks (13) independently. Finite computation is falsification pressure, not proof.

The rule depends on the ambient linear order and on discarding multiplicity; neither feature is invariant under arbitrary relabelling. We do not treat random updates, asymptotic limit shapes, or a classification of analogous multiset maps. The owner search was bounded, and the elementary nature of the rank subtraction leaves a live ownership risk. The status is therefore OWNER_AMBER / HOLD_EXTERNAL; no circulation or priority claim is authorized.

Data availability. No external data were used. The exact verifier and canonical transcript accompany the source.

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